

◆ PIT WALL ENGINEER

MOTORSPORT ENGINEER ACADEMY • SILVER TIER MASTERCLASS

FROM RAW TELEMETRY TO STRATEGIC DECISIONS

Seeing the Race Like an Engineer

Marc Guitart Frescó • Creator, Pit Wall Engineer

IN COLLABORATION WITH



WHERE THIS STARTED

THE MOST IMPORTANT STORIES IN F1 ARE THE ONES YOU NEVER SEE.

You see the overtake. Not the four laps of tyre data that made it possible.

You see the pit stop. Not the calculations behind it.

Ever since I started watching Formula 1, I was fascinated by the people making those decisions, not just the drivers. The engineers on the pit wall, turning data into decisions that define races.

This talk is about seeing what they see.

TEAM RADIO • 2024

"Leave me alone, I know what
I'm doing."

"It's hammer time."

THE PATH HERE

01 FIB · UPC

Computer Science in Software Engineering, Barcelona

02 KAFKA THESIS

BSc thesis on adaptive real time telemetry prediction, later a research paper

03 FERMAT

AI Systems Engineer Lead

THE DATA IS NOT THE PROBLEM

50K+

data points captured every session, the same numbers
drivers and engineers use to find every tenth of pace

FANS WATCH THE RACE.
ENGINEERS READ THE DATA THAT WINS IT.

RAW SIGNAL VS. STRATEGIC SIGNAL

WHAT TELEMETRY GIVES YOU

Timestamps and sector splits

Lap times, gaps to the car ahead

Tyre compound flags

Pit stop timestamps

WHAT ENGINEERS NEED TO KNOW

Is the undercut actually working?

How close is this tyre to falling off a cliff?

Did that stop gain time or lose it?

How much of this race is random?

WHERE MOST TOOLS STOP

A dashboard full of numbers is not the same as knowing what to do next.



Pit Wall Engineer is built to close this gap, automatically.

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IF IT DOESN'T ANSWER A STRATEGIC QUESTION, IT'S NOT IN THE MVP.

Product principle · Pit Wall Engineer

STRATEGIC QUESTIONS, NOT METRICS

Every module maps to one question a race engineer actually asks.

TRUE_PACE Who is actually fastest, once fuel and traffic are stripped out?

TYRE_DEG When does this tyre stop being competitive?

PIT_IMPACT Did that pit stop gain time or lose it?

CHAOS_INDEX How much of the result was randomness, not pace?

DECISIONS Which calls actually changed the outcome?

BEYOND THE MVP

The five questions above are the core loop. Eight more modules layer on top.

RACE_PHASE	Segments the race into 10 phases, SC, VSC, weather, pit windows, so no lap is mislabelled	RACE_DNA	An 8-point fingerprint of the race: tyre management, chaos, incident density, and more
DRS_TRAINS	Detects trains of 3+ cars stuck within a second, and how long they hold together	WX_CROSSOVER	Finds the wet/dry crossover window and grades each driver's call: early, on time, or late
CLEAN_AIR_VALUE	Measures how much of a driver's pace came from running in clean air	ENGINEER_NOTES	Automated qualitative read of the race, the same context that grounds the AI chat
RACE_ENGINEER_CHAT	Q&A grounded in the session's own analysis, never raw telemetry	TELEMETRY_REPLAY	FastF1-powered lap replay, speed, throttle, brake, synced to an animated circuit map

SYSTEM OVERVIEW



NORMALIZATION AND METRICS

Raw responses don't arrive aligned. Timestamps drift, laps overlap, drivers reorder. Everything is normalized onto a common lap and driver index before a single metric is computed.

TRUE_PACE

Lap time normalized for fuel load and traffic

DEGRADATION_SLOPE

Time lost per lap across a tyre stint

PIT_DELTA

Time gained or lost by a pit stop, against a modeled baseline

CHAOS_INDEX

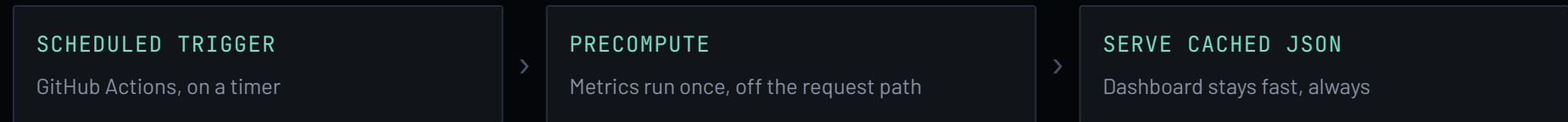
A composite volatility score for the whole race

PEAK_CHAOS_LAP

The single lap where that volatility spiked hardest

SERVING THE DASHBOARD

Telemetry-derived metrics are precomputed, not calculated on request.



The result is what you're about to see.

◆ CASE STUDY

2024 BRAZILIAN GRAND PRIX

Walking through the dashboard, panel by panel.

THE DASHBOARD, AT A GLANCE

TRUE_PACE

Reveals who was fastest once fuel and traffic are stripped away.

TYRE_DEG

Flags exactly when a stint's tyre fell off the pace.

PIT_IMPACT

Separates pit stops that gained time from ones that lost it.

CHAOS_INDEX

Scores how much of the result came down to chance, and flags peak_chaos_lap.

DECISIONS

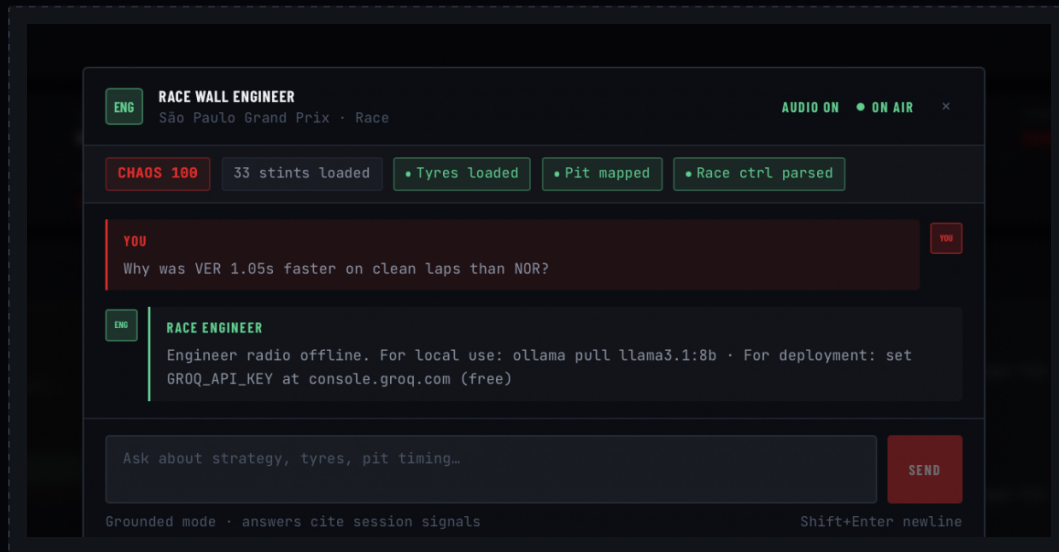
Ranks the five calls that actually changed the outcome.

◆ LIVE DEMO

NOW, LET'S LOOK AT THE REAL THING

pitwallengineer.com

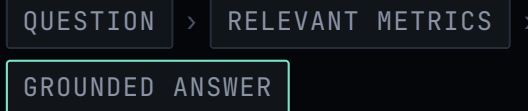
RACE ENGINEER DISCUSSION



ASK THE RACE ENGINEER

Ask a question in plain language, like why the undercut worked on lap 34. The system pulls only the precomputed metrics relevant to that question, then asks the model to explain them in words.

The model never touches raw telemetry, and it never does the math itself.



CONFIDENCE AND EXCLUSION LOG

ILLUSTRATIVE EXAMPLE

confidence: 0.82

```
exclusion_log: [  
  "lap 34 telemetry gap",  
  "SC period 2 excluded from pace calc"  
]
```

Confidence measures how complete the underlying metrics were, not how sure the model feels.

Every generated note carries its own confidence score and a log of what data was too incomplete to use. It reads like an engineer hedging a call, not a model asserting certainty it doesn't have.

LESSON: MODULE-LEVEL IMPORTS AND OOM

BEFORE

```
metrics.py

import pandas as pd # module top
```

Heavy libraries loaded at module import time. Every cold start paid the full memory cost before serving a single request.

AFTER

```
metrics.py

def compute():
    import pandas as pd
```

Imports moved inside the functions that need them. Memory is paid only when that code path actually runs.

LESSON: CACHING AND ABSOLUTE PATHS

STATIC CACHING VIA GITHUB ACTIONS

```
metrics.yml

on:
  schedule:
    - cron: "0 */6 * * *"
```

Metrics are precomputed on a schedule by a workflow, not on every dashboard request. The API serves cached JSON. The expensive computation happens once, off the request path.

ABSOLUTE PATHS IN PRODUCTION

```
config.py

DATA_DIR = os.environ["DATA_ROOT"]
```

Relative paths resolved fine locally and broke silently once deployed, depending on the process's working directory. Every file reference now resolves from an absolute, environment-defined root.

WHAT'S NEXT

- ◆ After-race analysis, a full breakdown once the race ends
- ◆ Live data reproduction, to follow the race as it happens
- ◆ Post-race notifications, so you can review it like a driver would with their race engineer
- ◆ A better reasoning model for the Pit Wall Engineer LLM
- ◆ New metrics, driven by member feedback

QUESTIONS

pitwallengineer.com

marc.guitart.fresco@gmail.com

<https://www.linkedin.com/in/marc-guitart-fresco/>

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